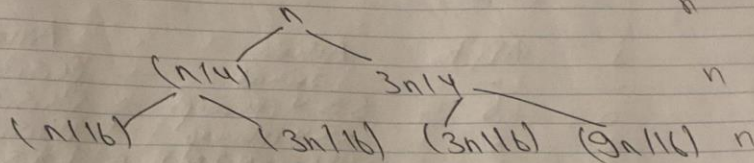


Homework 3

2.

$$T(n) = T(n/4) + T(3n/4) + O(\log n)$$



$$\text{left path} = \left(\frac{1}{4}\right)^k n = 1, k = \log_4 n$$

$$\text{right path} = \left(\frac{3}{4}\right)^k n = 1, k = \log_{4/3} n$$

$$\text{height} = O(\log n)$$

$$T(n) = n \cdot \log n$$

$$T(n) = O(n \log n)$$

Homework 3

Q. 2 a

2 a) $T(n) = 4T(n/2) + n^4 + 300$

$$T(n) = aT(n/b) + f(n)$$

$$a=4, b=2, f(n)=n^4$$

$$n \log_2 4 = n^2$$

$$f(n) = n^4$$

$$n^4 > n^2$$

$$T(n) = \Theta(n^4)$$

⑥ $T(n) = 4T(n/2) + (15)n^3 + 50n + 240$

$$a=4 \quad b=2 \quad f(n)=n^3$$

$$f(n) = \frac{n \cdot n^2}{n^3}$$

$$n^3 > n^2$$

$$T(n) = \Theta(n^3)$$

Q.2

$$c. T(n) = 5T(n/2) + 8n^2$$

$$a=5 \quad b=2 \quad f(n)=n^2$$

$$n \log_2 5 = n^{2.3}$$

$$f(n) = n^2$$

$$n^2 < n^{2.3}$$

$$T(n) = O(n \log_2 5)$$

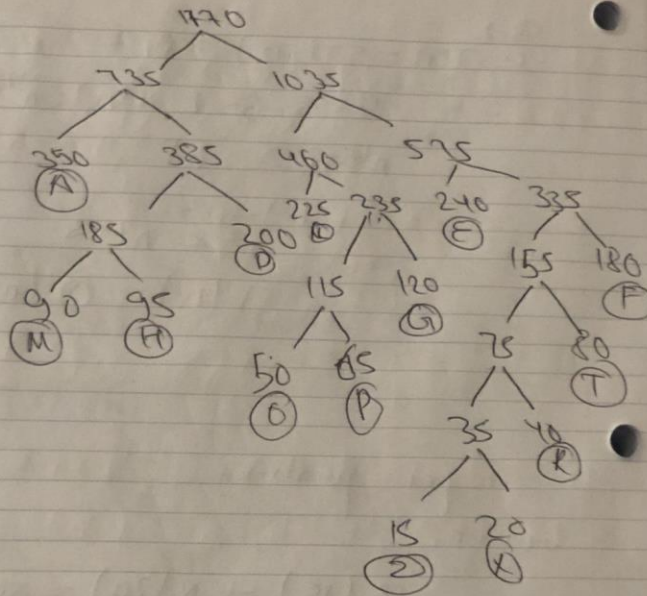
3. As per question

Given,

$$2(15) + x(20) = 35$$

$$x(40) + y(50) = 90$$

③



So,
we get,

$$A = 350 \times 2 = 700$$

$$E = 240 \times 3 = 720$$

$$C = 225 \times 3 = 675$$

$$D = 200 \times 3 = 600$$

$$F = 180 \times 4 = 720$$

$$G = 120 \times 4 = 480$$

$$H = 95 \times 4 = 380$$

$$M = 90 \times 4 = 360$$

$$Q = 50 \times 5 = 250$$

$$P = 65 \times 5 = 325$$

$$T = 80 \times 5 = 400$$

$$R = 40 \times 6 = 240$$

$$X = 20 \times 7 = 140$$

$$Z = 15 \times 7 = 105$$

$$P_{\text{sum}} = 6095$$

$$ABL = \frac{6095}{1770} = 3.44$$

$$\text{Huffman ABL} = 3.44$$

fixed length ABL 4 bits per letter

4.

Initial state

$S = S_4$

PQ = $S_5: 0, 2: \infty, 3: \infty, 4: \infty, 5: \infty, 6: \infty, 7: \infty, 1: \infty$

* (S10)

$S = S_8$

key operate from S

Node 2 : $\infty \rightarrow 12$

Node 6 : $\infty \rightarrow 11$

Node 7 : $\infty \rightarrow 13$

PQ : $S_6: 11, 2: 12, 7: 13$

* delmin 6/11

$S = S_5, 6$

decrease key op

Node 3 : $\infty \rightarrow 11 + 19 = 30$

Node 5 : $\infty \rightarrow 11 + 14 = 25$

Node 7 : $13 \rightarrow 13$

* delmin 2/12

$S = S_5, 6, 2$

decrease key operator

Node 3 : 30

* delmin 7/13

$S = S_5, 6, 2, 7$

decrease key operator.

Node 5 : 25 + 25

Node 4 : $\infty = 38$

* delmin 5

$S = \{5, 7, 8, 9\}$
decrease key op

Node 3: $30 \rightarrow 30$

Node 4: $\infty \rightarrow 35$

Node 4: $38 \rightarrow 38$

* delmi 3

decrease key for node 4
 $38 \rightarrow \min(38, 35+5) = 38$

Shortest distance to $t = 38$

Shortest Path = $S \rightarrow 7 \rightarrow t$

set $S = \{5, 2, 3, 4, 5, 6, 7, t\}$

5

(a) Pre order:

120, 42, 7, 2, 32, 24, 37, 40, 42

(b) Inorder:

2, 7, 24, 32, 40, 37, 42, 120, 42

(c) Postorder:

2, 24, 40, 37, 32, 7, 42, 42, 120

⑥

50

50 — 60

50
40 — 60

50
40 — 60
20

40
20 — 50
10 — 60

50
40 — 60
20
10

40
20 — 50
10 — 35 — 60

Imbalance at root: So
right rotation
at 50

⑦

[10, 20, 5, 6, 1, 8, 9, 30, 40, 50]

$$\text{index} = \frac{n}{2} - 1 = 4$$

Index 4 50 > 1
10, 20, 5, 6, 50, 8, 9, 30, 40, 1

Index 3 10, 20, 5, 40, 50, 8, 9, 30, 6, 1

Index 2 10, 20, 9, 40, 50, 8, 5, 30, 6, 1

Index 1 10, 50, 9, 40, 20, 8, 5, 30, 6, 1

Index 0 50, 10, 9, 40, 20, 8, 5, 30, 6, 1

⑧

- 50, 40, 9, 30, 20, 8, 5, 10, 6, 1
- 40, 30, 9, 10, 20, 8, 5, 1, 6 | 50
- 30, 20, 9, 10, 6, 8, 5, 1 | 40, 50
- 20, 10, 9, 1, 6, 8, 5 | 30, 40, 50
- 10, 6, 9, 1, 5, 8 | 20, 30, 40, 50
- 9, 6, 8, 1, 5 | 10, 20, 30, 40, 50
- 8, 6, 5, 1 | 9, 10, 20, 30, 40, 50
- 6, 1, 5 | 8, 9, 10, 20, 30, 40, 50
- 5, 1 | 6, 8, 9, 10, 20, 30, 40, 50
- 1 | 5, 6, 8, 9, 10, 20, 30, 40, 50
1, 5, 6, 8, 9, 10, 20, 30, 40, 50